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Construction Joist and Joist Assembly

Abstract

A roof and floor framing assembly are described, using a steel construction joist having a 16-inch height, for building single-family homes in compliance with Type 1A non-combustible definition in Chapter 6 of the California Building Code (CBC). The joist assembly includes a plurality of steel joists bolted directly to opposing load-bearing reinforced concrete walls using knife plates, and spaced approximately 96 inches apart for most applications. A plurality of insulated steel roof panels or floor panels are arranged and connected perpendicular to and covering the plurality of steel joists. In roof construction, the plurality of steel joists are arranged perpendicular to the direction of slope of the roof.

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Background/Summary

RELATED APPLICATION [0001] The present disclosure claims the filing priority of U.S. Provisional Application No. 63/560,528, titled “Construction Beam and Beam Assembly” filed on Mar. 1, 2024. The '528 application is also hereby incorporated by reference.

TECHNICAL FIELD OF THE INVENTION

[0002] The present invention relates to a construction joist, joist assembly, and roof and floor construction for a building. More specifically, the invention relates to a joist construction and assembly for constructing components of a non-combustible building structure, such as a house.

BACKGROUND OF THE INVENTION

[0003] Wildfires are a major cause of home damage in many areas of the U.S., particularly in the western states. According to an article in Scientific American (“Wildfires Threaten More Homes and People in the U.S. Than Ever Before”, by Stephanie Pappas, *Scientific American* (a division of Springer Nature America, Inc.), Nov. 9, 2023), the increase in home damage is the result of an increased expansion of building into “wildland urban interface,” or WUI, which it describes as anywhere houses butt up against natural areas.

[0004] Grassland fires have been responsible for some of the most devastating losses in recent years. Most recently, from Jan. 7 to Jan. 31, 2025, a series of eight destructive wildfires affected the Los Angeles metropolitan area and San Diego County in California. In Los Angeles County, a report from the UCLA Anderson Forecast estimates that the two largest wildfires from that event—i.e., the Palisades and Eaton fires—caused total property and capital losses ranging between \$95 and \$164 billion, with insured losses estimated at \$75 billion. There have been 29 deaths attributed to those fires. In the fire that leveled the Hawaiian town of Lahaina (August 2023), strong winds aided the fire that tore through hillsides of invasive grasses—and became an urban conflagration that destroyed more than 2,000 structures and killed at least 99 people. In December 2021, the Marshall Fire in Colorado's Boulder County destroyed more than 1,000 buildings and killed two people as it jumped from grasslands to suburban neighborhoods.

[0005] As illustrated in the graph of FIG. 1, both the area burned (grey area) and the number of homes (black bars) within the perimeter of such fires has increased significantly since 1990.

[0006] As the desire for a wildland urban interface for homes grows, so does the risk of fire damage. And, because the typical family home is built using a high percentage of combustible materials, such as wood, wallboard, and the like, the threat of complete loss due to a fire is very high. Accordingly, there needs to be a way to build houses close to nature, while minimizing the threat of fire damage and calamity.

[0007] Home builders and home buyers have turned to looking for homes built with less combustible materials. At present, there are homes that are built with steel framing (not very common) and homes that are built with Insulated Concrete Forms, but they also have to use wood framing, particularly on interior load-bearing walls.

[0008] California building codes, which are probably the most restrictive of any in the United States and possibly in the World, defines “Type 1A Non-Combustible” homes in Chapter 6 of the California Building Code 2019 (and International Building Code (IBC) 2018). However, there are no Type I-A single family homes being built. The regulations are very strict and present a difficult requirement to meet. For example, structural walls in a home need to have a minimum 3-hour fire rating, floors a 2-hour rating, and a roof must have a minimum 1.5-hour fire rating. These are very difficult requirements to meet cost-effectively, and literally impossible using wood materials.

[0009] Accordingly, a system and method are needed which allows for cost-effective compliance with strict building codes to achieve Type 1A Non-Combustible homes for those living in areas at risk of wildfires. Until the invention of the present application, these and other problems in the

prior art went either unnoticed or unsolved by those skilled in the art. The present inventions provide an effective construction joist and building method which achieves multiple goals, including fire resistance, without sacrificing design, style, or affordability.

SUMMARY OF THE INVENTION

[0010] There is disclosed herein an improved construction method, including a unique roof and floor framing assembly which avoid the disadvantages of prior construction framing assemblies while affording additional structural and operating advantages.

[0011] Generally speaking, the steel construction joist has a 16-inch height and is used for building single-family homes in compliance with Type 1A non-combustible definition in Chapter 6 of the California Building Code (CBC).

[0012] The roof and floor framing assemblies include a plurality of steel joists bolted directly to opposing load-bearing walls, and spaced approximately 96 inches apart, a plurality of stiffener flanges may be used on the joists, as well as a plurality of steel braces arranged to connect adjacent joists. A plurality of insulated steel panels perpendicular to and covering the plurality of steel joists and bracing completes the assembly construction. A spacing of 4 to 8 feet between joists provides substantial room for the addition of mechanicals and lighting.

[0013] In specific embodiments, the joists may span a distance of 15 to 32 feet between load-bearing support walls. Finally, the plurality of steel joists is arranged perpendicular to the direction of slope for a roof.

[0014] These and other aspects of the invention may be understood more readily from the following description and the appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] For the purpose of facilitating an understanding of the subject matter sought to be protected, there are illustrated in the accompanying drawings, embodiments thereof, from an inspection of which, when considered in connection with the following description, the subject matter sought to be protected, its construction and operation, and many of its advantages should be readily understood and appreciated.

[0016] FIG. 1 is a graph reproduced from “Rising Wildfire Risk to Houses in the United States, Especially in Grasslands and Shrublands” by Volker C. Radeloff et al., in *SCIENCE*, vol. 382 (Nov. 9, 2023), showing the impact of wildfires;

[0017] FIG. 2 is a perspective view of an embodiment of the disclosed joists and framing assembly as they would be used for constructing a Type 1A non-combustible home;

[0018] FIG. 3A is a front view of an embodiment of a knife plate used to bolt the disclosed joist to loadbearing outer walls;

[0019] FIG. 3B is a side view of the knife plate of FIG. 3A;

[0020] FIG. 3C is a top view of the knife plate of FIG. 3A;

[0021] FIG. 4 is a cut-away side view of an embodiment of the disclosed joist bolted to an outer support wall using an embodiment of the knife plate of FIG. 3A;

[0022] FIG. 5A is a front view of an embodiment of a knife plate used to bolt the disclosed brace to loadbearing outer walls;

[0023] FIG. 5B is a side view of the knife plate of FIG. 5A;

[0024] FIG. 5C is a top view of the knife plate of FIG. 5A;

[0025] FIG. 6 is a cut-away side view of an embodiment of the disclosed brace bolted to an outer support wall using an embodiment of the knife plate of FIG. 5A;

[0026] FIGS. 7 and 8 are views showing placement of an embodiment of a plurality of roof joists between two load-bearing walls in a home construction;

[0027] FIGS. **9** and **10** are views showing placement of an embodiment of a plurality of bracing used in a construction similar to FIG. **2**;

[0028] FIG. **11** is a cut-away view of a construction showing embodiments of the disclosed floor assembly and roof assembly;

[0029] FIGS. **12** and **13** are views illustrating placement of lighting and mechanicals in a construction similar to FIG. **2**;

[0030] FIGS. **14** through **16** are views illustrating placement of reinforcement to a construction similar to FIG. **2**; and

[0031] FIGS. **17** through **20** are views illustrating placement of an embodiment of a plurality of insulated roof panels to a construction similar to FIG. **2**.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

[0032] While this invention is susceptible of embodiments in many different forms, there is shown in the drawings and will herein be described in detail at least one preferred embodiment of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to any of the specific embodiments illustrated. The use of a “joist” is well-known in the construction industry and, in some instances, may also be referred to as a “beam.” Similarly, “braces” or “bracing” as used herein may also be referred to in the industry as “purlins.”

[0033] Referring to FIGS. **2-20**, there is illustrated a housing construction joist and assembly, generally designated by numerals **10** and **12**, respectively. Roof joists, referenced as **10A**, and floor joists, referenced as **10B**, are otherwise indistinguishable and often referenced collectively as joist **10**. Likewise, a roof assembly **12A** and floor assembly **12B**, are often collectively referred to as assembly **12**. The disclosed and illustrated joist **10** and assembly **12** is for constructing single-family Type 1A Non-Combustible homes in accordance with Chapter 6 of the California Building Code 2019 (and IBC 2018). However, while all the embodiments illustrated are directed to roof and floor construction for a single-family home, it should be understood that the principles of the invention can be more broadly applied to most dwellings, as well as other types of buildings.

[0034] With reference to FIG. **2**, joist **10** is constructed of a steel material, preferably a steel material having a high-tensile strength and outer walls **20** are comprised of a reinforced concrete. Preferably, each joist **10** is approximately 16 inches in height and spans from a first loadbearing reinforced concrete wall **22** to a second loadbearing reinforced concrete wall **24**. Joists **10** are preferably bolted directly to each outer concrete wall **20** and spaced apart a distance in the range of from 4 feet (48 inches) to 8 feet (96 inches). The actual spacing between adjacent joists **10** used in each construction should be determined based on load-bearing requirements of each joist **10**, as would be known in the art. In the preferred embodiment, each joist **10** can span up to 32 feet—from a first to a second loadbearing concrete wall, **22** and **24**, respectively. The most common span expected for each joist **10** falls in the range of 15 to 28 feet, more specifically in the range of 20 to 26 feet. Of course, shorter spans below 15 feet are contemplated where the loadbearing walls are a shorter distance apart, for example. These long spans between support walls allow interior walls to be constructed from lightweight metals instead of reinforced concrete, which saves considerable time and money in construction.

[0035] Referring to FIGS. **3-8**, the joists **10** are bolted directly to an outer concrete wall **20** using unique knife plates **40A** and **40B**. The knife plate **40A**, as illustrated in FIGS. **3A-3C**, is comprised of four half-inch studs **42** extending perpendicularly from a first side of a base plate **44**, and a flange **46** extending from a second (opposite) side of the base plate **44**. The studs **42** of the knife plate **40A** are embedded in an outer concrete wall **20**, as shown in FIG. **4**, to allow a joist **10A** to be bolted to the flange **46**. Similarly, a brace knife plate **40B** is used for securing braces **30** to outer concrete walls **20** as well. The knife plate **40B**, as shown in FIGS. **5A-5C**, is comprised of four half-inch studs **42** extending perpendicularly from a first side of a base plate **44**, and a flange **46** extending from a second (opposite) side of the base plate **44**. The studs **42** of the knife plate **40B**

are embedded in the outer concrete wall **20**, as shown in FIG. **6**, to allow braces **30** to be bolted to the flange **46**.

[0036] As shown in FIGS. **7-10**, each roof joist **10A** may be constructed with stiffener flanges **28** to provide rotational stability when used for roof slopes up to 4:12. Further, as shown in FIG. **11**, roof joists **10A** are arranged to run perpendicular instead of parallel to the direction of roof slope. Metal braces **30** may also be positioned to connect between adjacent roof joists **10A** (as well as floor joists **10B**) in most applications.

[0037] As illustrated best in FIGS. **12** and **13**, the joist assembly **12** provides adequate room for mechanical and lighting installation, as no batting is required. The joist **10** enables framing and roofing of a house in about one-third of the time that it normally takes using wood materials. This labor efficiency provides a direct cost savings. As previously noted, the preferred eight-foot spacing between joists **10** and no batt insulation means that there is plenty of clear space for HVAC, lighting fixtures, plumbing, electrical wiring, and fire sprinklers-unlike wood framing having 12"-24" joist spacing and fiberglass batt requirements causing all the trades to compete with one another for adequate space. The result of using wood framing by comparison is messy and time consuming.

[0038] Steel structural roof panels **32** are shown completing the construction in FIGS. **17-20**. The use of insulated standing seam metal roof panels **32** provides further resistance to lateral forces, thereby eliminating the need for structural sheathing. In turn, the elimination of structural sheathing enables far greater cost-effective construction of an entirely wood-free residential roofing assembly **12A**. Where a floor assembly **12B** is constructed, composite decking **34** is preferably used to complete the construction. The composite decking **34** is comprised of steel sheeting with a topping of 3,000 psi lightweight concrete. Obviously, other floor materials may be suitable, so long as they meet requirements for non-combustible construction materials.

[0039] The disclosed invention further includes constructing a non-combustible single-family home in compliance with the California code. Beginning with the erection of outer walls **20**, these are made from reinforced concrete. These outer walls **20** become the primary load-bearing structures for the home. At each level or story, a floor assembly **12B** is built in accordance with the above description. When constructing a home of multiple stories (i.e., two or more stories), it will be necessary to build up the outer walls **20** after each of the multiple floor assemblies **12B** is built. A final outer wall construction supports the roof assembly **12A**, as shown and described.

[0040] In this manner, the joist assembly **12** helps realize a non-combustible minimum 1.5-hour fire rated roof assembly **12A** and a 2-hour fire rated floor assembly **12B** for a single-family home. The joist assembly **12** eliminates the need for structural sheathing, where plywood is the only feasible sheathing option for single-family homes. Without the disclosed joist **10** and joist assembly **12**, building a Type I-A home would be so expensive that it would not be feasible.

[0041] The matter set forth in the foregoing description and accompanying drawings is offered by way of illustration only and not as a limitation. While particular embodiments have been shown and described, it will be apparent to those skilled in the art that changes and modifications may be made without departing from the broader aspects of applicants' contribution. The actual scope of the protection sought is intended to be defined in the following claims when viewed in their proper perspective based on the prior art.

Claims

1. A non-combustible roof assembly for a residential construction having a roof and a plurality of reinforced concrete outer walls, the roof assembly comprising: a framing assembly comprising: a plurality of steel joists, each perpendicular to a direction of slope of the roof and spaced apart a first distance from one another, wherein each of the plurality of steel joists is secured to and spans a second distance of at least 15 feet between two of the plurality of reinforced concrete outer walls; a

plurality of steel braces arranged to connect to and between adjacent steel joists from the plurality of steel joists; and a plurality of steel structural roof panels secured and perpendicular to the plurality of steel joists.

2. The non-combustible roof assembly of claim 1, wherein the first distance is in a range of from 4 to 8 feet.
3. The non-combustible roof assembly of claim 2, wherein the first distance is 6 feet.
4. The non-combustible roof assembly of claim 1, wherein the second distance is at least 20 feet.
5. The non-combustible roof assembly of claim 1, wherein each of the plurality of steel beams comprises a plurality of stiffener flanges to provide rotational stability.
6. The non-combustible roof assembly of claim 4, wherein the second distance is at least 26 feet.
7. The non-combustible roof assembly of claim 1, wherein the roof assembly complies with Type 1A non-combustible definition in Chapter 6 of the California Building Code (CBC).
8. A floor assembly for a residential construction having at least one floor and a plurality of reinforced concrete outer walls, the floor assembly comprising: a joist assembly comprising: a plurality of steel joists, each spaced apart a first distance from one another, wherein each of the plurality of steel joists is secured to and spans a second distance of at least 15 feet between two of the plurality of reinforced concrete outer walls; a plurality of steel braces arranged to connect to and between adjacent steel joists from the plurality of steel joists; and a plurality of steel structural floor decking secured and perpendicular to the plurality of steel joists.
9. The non-combustible floor assembly of claim 8, wherein the first distance is in a range of from 4 to 8 feet.
10. The non-combustible floor assembly of claim 9, wherein the first distance is 8 feet.
11. The non-combustible floor assembly of claim 8, wherein the second distance is at least 20 feet.
12. The non-combustible floor assembly of claim 8, wherein each of the plurality of steel joists comprises a plurality of stiffener flanges to provide rotational stability.
13. The non-combustible floor assembly of claim 11, wherein the second distance is at least 26 feet.
14. The non-combustible floor assembly of claim 8, wherein the roof assembly complies with Type 1A non-combustible definition in Chapter 6 of the California Building Code (CBC).
15. A method for constructing a non-combustible residential home comprising: erecting a plurality of reinforced concrete outer walls; building a floor assembly for each story of the non-combustible residential home, each floor assembly comprising: a first joist assembly comprising: a plurality of steel joists, each spaced apart a first distance from one another, wherein each of the plurality of steel joists is secured to and spans a second distance of at least 15 feet between two of the plurality of reinforced concrete outer walls; a plurality of steel braces arranged to connect to and between adjacent steel joists from the plurality of steel joists; and a plurality of steel structural floor decking secured and perpendicular to the plurality of steel joists; erecting a non-combustible roof assembly comprising: a second joist assembly comprising: a second plurality of steel joists, each perpendicular to a direction of slope of the roof and spaced apart a third distance from one another, wherein each of the plurality of steel joists is secured to and spans a fourth distance of at least 15 feet between two of the plurality of reinforced concrete outer walls; a second plurality of steel braces arranged to connect to and between adjacent steel joists from the second plurality of steel joists; and a plurality of steel structural roof panels secured and perpendicular to the second plurality of steel joists.
16. The method of claim 15, wherein the first distance is in a range of from 4 to 8 feet.
17. The method of claim 16, wherein the first distance is 8 feet.
18. The method of claim 15, wherein the second distance is at least 20 feet.
19. The method of claim 15, wherein each of the plurality of steel joists comprises a plurality of stiffener flanges to provide rotational stability.
20. The method of claim 18, wherein the second distance is at least 26 feet.

21. The method of claim 15, wherein the non-combustible residential home complies with Type 1A non-combustible definition in Chapter 6 of the California Building Code (CBC).
